

Michelle Star

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Education

University of Pittsburgh,

Apr. 2026

Honors College, School of Computing and Information (*summa cum laude*)

Bachelor of Philosophy in Data Science, *Specialization: Data Analytics* | Minors: Computer Science, Theater Arts

Thesis: “Simulating the User: Multi-Turn Evaluations of Conversational AI Support Across Demographic Contexts”

Faculty Mentor: Dr. Yu-Ru Lin

Honors: David C. Frederick, *Honors Distinction*

Neapolis University Pafos Paphos, Cyprus | Study Abroad Program in Sociology

May 2025

Taylor Allderdice High School (*Valedictorian*) Pittsburgh, PA

Jun. 2022

Research Interests

Undergraduate researcher with a technical and interdisciplinary background in data science, human-computer interaction (HCI), and accessible technology. Currently completing a BPhil thesis on conversational AI bias across demographics and leading user research for my assistive tech startup. Recent work includes developing an award-winning LLM-based accessibility platform for wheelchair users. Seeking PhD opportunities to advance research in socially responsible, user-centered computing.

Research Experience

Undergraduate Researcher, BPhil Thesis, University of Pittsburgh

2025–Present

- Investigating support variance and emergent bias in LLMs across demographic subcommunities on Reddit.
- Using NLP and multi-turn simulations to analyze linguistic sentiment and support quality to reveal discrepancies.
- Conducting original research to be defended before a committee of faculty and experts in the field.

Volunteer Research Assistant, CivicWatch, PICSO Lab

2025–Present

- Extracted and classified political discourse on social media for legislative transparency research.
 - Worked on NLP modeling and front-end design to improve civic accessibility and reduce misinformation.
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Internships

Data Science Capstone, Latitude AI (Ford Subdivision)

2025–Present

- Designed and built a data pipeline to aggregate and analyze road work zone data utilizing open-source data from across the U.S., supporting the safety of autonomous vehicle navigation.
- Processed and standardized diverse datasets to provide real-time construction zone awareness for integration into Ford’s self-driving systems.

UX/UI Designer, Osher Lifelong Learning Institute

2025–Present

- Leading a website redesign initiative to improve modernity, accessibility, and usability for an older adult audience.
 - Conducted user experience research to identify pain points in the current interface, focusing on screen reader compatibility and intuitive information flow.
 - Implementing accessibility best practices (WCAG) to ensure content is perceivable, navigable, and user-friendly for individuals with visual or cognitive impairments.
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Leadership & Mentorship Experience

Peer Reviewer, Pittsburgh Undergraduate Review

2025–Present

- Selected to review interdisciplinary undergraduate research submissions.
- Completed training led by faculty on research evaluation and academic peer review.
- Provided written feedback on research clarity, methodology, and contribution.

Appointed Student Representative, Academic Council, School of Computing and Information

2025–Present

- Selected by the Dean as one of two undergraduates to represent students in school-wide curricular governance.
- Participated in monthly council meetings to review, deliberate, and vote on proposed changes to academic programs, degree requirements, and policies.

Mentor, Women in Computer Science Club & CS Club, University of Pittsburgh 2025–Present

- Mentored 5 undergraduates in academic success, career exploration, and navigating the tech field.
- Guided on course planning, research/internship opportunities, hackathons, and professional communication.
- Fostering confidence and retention of women in tech by sharing resources and facilitating networking opportunities.

Program Director, YMCA Camp Kon-O-Kwee Spencer 2021-2025

- Directed experiential learning programs for 560+ youth campers, overseeing two departments.
- Mentored and evaluated a team of 15 staff, emphasizing skill development, iterative performance feedback.
- Partnered with healthcare professionals to co-develop accessible camp experiences for children with Hemophilia and Charcot-Marie-Tooth, incorporating accessibility principles into educational design.

Extended Day Counselor, St. Edmund's Academy 2023-Present

- Independently facilitated structured after-school learning for 40+ students.
- Acted as primary educator in the absence of teachers, supported behavioral needs across a diverse student population.
- Provided individualized math tutoring and homework assistance to reinforce classroom learning.

Selected Projects

Team Leader, Wireless Innovation Hackathon for Accessibility - First Place Winner - (\$1,500 Prize) Oct. 2025

- Created a platform to simplify access to wheelchair-accessible vehicles (WAVs) using LLM-powered agents that contact transportation providers on users' behalf.
- Collaborated with a community member with Cerebral Palsy to ensure real-world relevance and usability.
- Built with Google Maps API integration, enabling route comparison and integrated WCAG-compliant UI features.
- Recognized for innovation, inclusive design, and societal impact by a cross-disciplinary panel of judges.

Automated Program Zone Form Jun. 2025

- Automated the assignment of 200+ campers to activities, reducing administrative workload by 10+ hours per week.
- Identified program inefficiency, improved fairness by systematically balancing camper preferences and staff capacity.

Three Rivers Auto Data Science Consultant Final Project Apr. 2025

- Conducted full analysis of 1,800+ historical car listings to identify key factors influencing used car prices, acting as Data Science consultant and delivering technical and non-technical reports for actionable business decisions.
- Built and compared multiple predictive models, including Linear Regression, Ridge, Lasso, Random Forests, GAM.

Prevalence of Heavy Drinking Survey Study Apr. 2025

- Developed questionnaire and cluster sampling method covering demographics, consequences, and perceptions.
- Created plans to minimize non-response bias (reminders, incentives, weighting) and control measurement error.

Pitt Challenge Hackathon - Winner of GoDaddy Registry Prize Sept. 2023

- Developed "No Nurses Left Behind," a social media platform aimed at combating medical professional burnout.
- Integrated an AI chatbot using Python to provide real-time emotional support.

Presentations

Higher Ed Student Spotlight, The Global Impact Forum (TGIF) 2025

- Selected to present AI accessibility project MobilU to an audience of 300+ global leaders across industries.
- The event featured networking, keynote speakers, and a showcase of interdisciplinary impact initiatives.

Awards & Scholarships: Academic Merit Scholarship	2022 - Present
Pittsburgh Public Scholar	2022 - Present
Dean's List - School of Computing and Information	2023 - Present
Portrait Master Award - La Roche College	2022

Skills: Programming Languages: R, Python, Java, SQL, JavaScript, HTML, Assembly

Tools & Environments: RStudio, Visual Studio, Git, Jupyter, Mars, Figma, Logism

Data Science Tools: tidyverse, ggplot2, caret, Pandas, Matplotlib, keyBERT, HuggingFace

Certifications: *Disabling Ableism: Allyship and Advocacy Badge* (SHRS, University of Pittsburgh)

Programming Foundations with JavaScript, HTML, and CSS (Duke University, [Coursera](#))